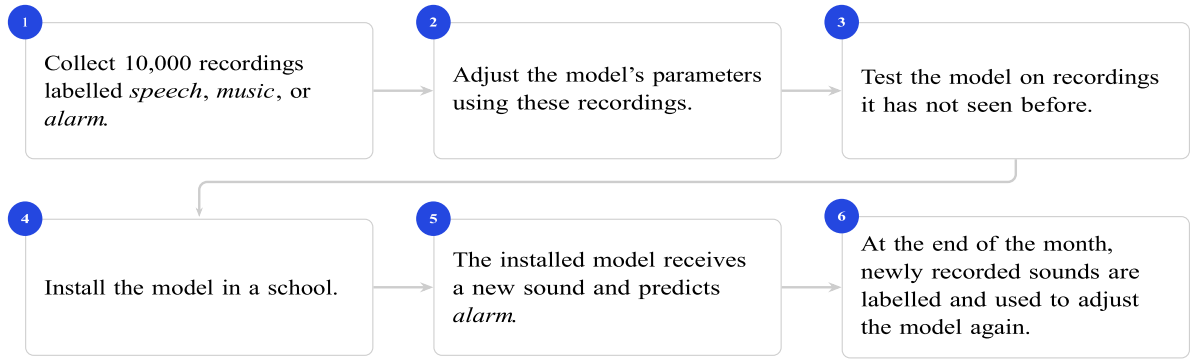
## Problem A Machine Learning Lifecycle (5 Points)

A sound-recognition system is developed through the six steps shown below.

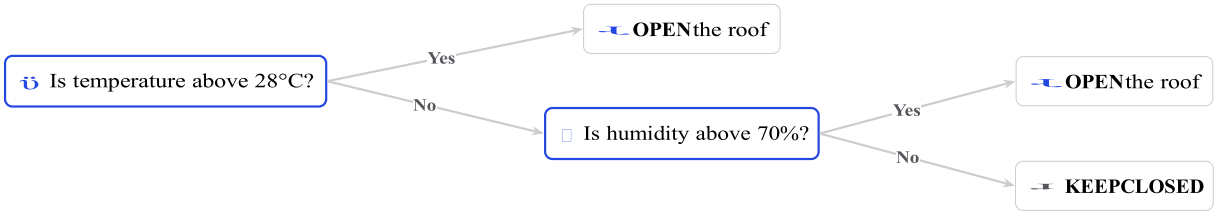



(a) Label each step as data collection, training, evaluation, deployment, or inference. A step may have more than one label.

(b) During which steps is the system actually learning? Explain what learning means here.

## Problem B The Greenhouse Tree (5 Points)

A greenhouse controls its roof with this decision tree:




(a) What action does the tree predict when the temperature is 26°C and the humidity is 68%? (b) A new $CO_2$ sensor is installed, and the manager begins logging the conditions together with what the roof should have done:

| Temperature | Humidity | $CO_2$ | Roof |
|---|---|---|---|
| 31°C | 60% | 700 ppm | OPEN |
| 26°C | 75% | 650 ppm | OPEN |
| 25°C | 60% | 800 ppm | KEEP CLOSED |
| 27°C | 68% | 1,100 ppm | KEEP CLOSED |
| 24°C | 55% | 1,400 ppm | OPEN |
| 22°C | 50% | 1,550 ppm | OPEN |

Adjust the decision tree so that it correctly models every observation in the log.

## Problem C To Fit or Not to Fit (5 Points)

A model predicts $y$ from $x$. The training data are:

| $x$ | 0 | 1 | 2 | 3 |
|---|---|---|---|---|
| $y$ | 1 | 3.2 | 4.8 | 7 |

Two models are proposed:

$$M_1(x) = 0.2x^3 - 0.9x^2 + 2.9x + 1, \qquad M_2(x) = 2x + 1.$$

Models are scored using

$$J = \sum (y - \hat{y})^2 + \lambda \sum_i a_i^2,$$

where the second sum runs over the model's coefficients of $x$, $x^2$, and $x^3$.

(a) For $\lambda = 1$, calculate the score $J$ for each model. Which model does the scoring rule select? (b) Which model fits the training data better? Explain why the score $J$ nevertheless selects the other model, and what the term $\lambda \sum_i a_i^2$ is for.

## Problem D The Price of Drift (5 Points)

A language model is trained to maximise a fixed reward so that its outputs better match desired qualities, such as being helpful and aligned with human preferences. Without a constraint, however, the model may drift far from its trusted starting point, the reference model, and learn to game the reward rather than produce genuinely good answers.[1]

Let $t \geq 0$ measure the drift from the reference model. Suppose the reward is $rt$ and the drift penalty is $\beta t^2$, with $r, \beta > 0$. The resulting loss is

$$L(t) = -rt + \beta t^2.$$

and training chooses the value of $t$ that minimizes it. In real systems, drift is typically measured using KL divergence; here we use $t^2$ for simplicity.

(a) Find the shift $t_*$ that minimizes the loss, and the loss value it reaches.

(b) What happens to $t_*$ as $\beta \to 0$, and as $\beta \to \infty$? Explain in one or two sentences what each case means for the language model.

(c) The reward estimate $r$ is noisy: the engineers only know that $0 < r \leq r^{\max}$. They require the shift to stay below a safety limit $T$ no matter what $r$ turns out to be. Show that this is guaranteed exactly when $\beta \geq \frac{r^{\max}}{2T}$.

[1] This is a simplified account of RLHF (reinforcement learning from human feedback). For an introduction, see e.g. https://huggingface.co/blog/rlhf.

(d) What does the term $\lambda \sum_i a^2_i$ from Problem C have in common with the penalty term $\beta t^2$ here?

## Problem E□ Farming Advice □5 Points)

An organisation wants to introduce a machine-learning system for farmers in rural Nepal. A farmer provides information about a crop, and the system recommends whether to irrigate, apply treatment, wait, or contact an expert. The organisation wants to deploy an off-the-shelf large language model, such as ChatGPT, Claude, or Gemini. Discuss three opportunities and three risks of introducing this system.

